

RIVIERE-DU-LOUP, QC, Canada

WMO: 717150

Lat: 47.8061N

Lon: 69.5486W

Elev: 481

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-7.0	-22.9	1.6	-10.1	-17.8	2.1	-5.7	20.9	23.3	18.4	20.1	6.4	190	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	79.4	67.6	76.6	66.2	74.1	64.8	70.0	76.4	68.2	74.1	66.4	72.2	8.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	103.6	73.9	65.9	97.0	72.2	64.0	90.7	70.3	34.4	76.4	32.8	74.2	31.4	72.5	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.8	15.4	13.6	-15.8	85.3	3.9	3.0	-18.6	87.5	-20.9	89.3	-23.1	91.0	-26.0	93.2		
			-16.6	73.7	3.9	2.1	-19.4	75.2	-21.6	76.5	-23.8	77.7	-26.6	79.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	11.3	14.2	23.5	35.9	48.7	58.3	64.1	62.6	54.6	42.9	31.7	19.4
	DBStd	19.98	11.58	10.58	9.71	7.03	7.14	6.79	5.52	5.59	6.87	7.25	8.36	9.93
	HDD50	5342	1200	1002	824	425	112	7	0	0	28	244	551	949
	HDD65	9594	1665	1422	1288	873	509	222	84	115	319	684	999	1414
	CDD50	1350	0	0	1	2	71	257	436	390	166	25	2	0
	CDD65	125	0	0	0	0	2	21	55	40	7	0	0	0
	CDH74	582	0	0	0	0	26	105	256	162	33	0	0	0
	CDH80	66	0	0	0	0	2	13	34	15	2	0	0	0
Wind	WSAvg	6.1	6.9	7.0	7.2	6.8	6.2	5.3	5.0	4.9	5.4	6.0	6.5	6.6
Precipitation	PrecAvg	38.5	2.7	2.5	2.8	2.9	3.5	3.3	3.9	3.9	3.4	3.4	3.0	3.7
	PrecMax	48.1	5.3	4.6	4.6	6.9	7.2	6.3	10.4	12.0	7.6	7.3	5.4	12.6
	PrecMin	29.8	0.7	1.3	1.2	1.1	1.1	1.0	0.8	1.4	2.0	0.9	1.1	1.3
	PrecStd	4.6	1.1	0.9	0.9	1.5	1.4	1.3	2.4	2.0	1.2	1.7	1.0	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	43.1	50.3	62.2	77.7	81.5	83.8	82.1	77.8	67.4	60.2	44.5
		MCWB	43.3	39.3	43.2	51.3	61.8	67.8	70.3	69.2	67.9	59.5	54.9	42.7
	2%	DB	37.6	36.2	43.2	54.8	71.0	76.9	79.8	78.1	73.5	63.1	51.6	38.3
		MCWB	36.1	33.5	37.7	45.2	58.3	65.7	68.5	67.2	65.1	56.8	47.2	36.5
	5%	DB	32.4	32.9	39.5	50.0	66.1	73.4	76.7	75.3	69.7	58.5	47.1	35.2
		MCWB	31.1	31.1	35.1	42.4	55.9	63.3	66.7	65.7	62.8	53.3	43.6	33.2
	10%	DB	27.9	29.9	36.3	46.1	61.5	70.3	73.9	72.7	66.3	54.1	42.7	32.7
		MCWB	26.5	28.2	32.8	40.1	52.3	61.5	65.3	64.5	60.3	49.7	39.5	31.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.6	40.4	45.7	53.6	65.0	71.3	73.4	71.3	69.6	61.4	56.0	43.3
		MCDB	44.6	42.3	48.5	59.4	74.4	79.1	80.9	78.5	75.9	66.1	58.3	44.6
	2%	WB	36.2	34.1	38.6	46.3	61.0	67.7	70.6	69.2	66.4	57.3	48.4	36.8
		MCDB	37.4	35.9	41.9	52.4	68.4	73.8	77.0	75.3	71.6	62.1	50.8	38.0
	5%	WB	31.1	31.3	35.5	43.3	56.7	65.3	68.6	67.5	63.7	54.0	43.7	33.6
		MCDB	32.4	32.7	38.8	49.1	63.7	71.3	74.4	73.4	68.9	58.1	46.6	34.9
	10%	WB	26.4	28.4	33.0	40.5	53.2	62.8	66.5	65.6	60.8	50.2	40.1	31.3
		MCDB	27.8	29.8	35.9	45.1	60.3	68.8	72.4	71.2	65.6	53.5	42.4	32.5
Mean Daily Temperature Range	5% DB	MDBR	13.6	13.6	13.0	13.0	16.5	16.7	15.9	15.8	15.1	11.7	10.1	11.0
		MCDBR	17.1	13.8	15.1	20.0	23.8	20.1	18.8	18.0	18.6	17.1	15.6	12.9
		MCWBR	17.1	12.7	12.3	13.8	14.3	12.6	10.9	10.4	12.7	12.8	13.8	12.8
	5% WB	MCDBR	17.3	13.2	13.7	17.9	21.5	17.7	17.1	16.5	18.3	16.1	15.4	12.9
		MCWBR	17.2	12.5	11.9	13.6	14.4	12.7	11.4	11.1	13.8	13.1	14.2	13.0
Clear-Sky Solar Irradiance	taub	0.267	0.275	0.312	0.354	0.360	0.378	0.405	0.385	0.361	0.328	0.294	0.271	
	taud	2.319	2.287	2.250	2.306	2.368	2.364	2.364	2.310	2.374	2.433	2.513	2.476	2.359
	Ebn at Noon	259	283	287	285	286	280	271	272	268	262	249	242	
	Edh at Noon	23	29	36	38	37	37	39	35	30	24	20	19	
All-Sky Solar Radiation	RadAvg	417	707	1087	1407	1610	1734	1722	1492	1154	675	369	282	
	RadStd	49	83	81	115	163	117	168	99	82	59	41	27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page